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Full Length Article

Hydroxyl speciation and water phases on sub-nanometer Ru_{10} clusters supported on anatase $TiO_2(101)$

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ABSTRACT

Ab initio thermodynamics of water on flat Ru(0001) predicts full dissociation to atomic O and H, with no stable hydroxyl, contradicting the experimental consensus that OH is an essential intermediate on Ru/TiO_2 hydrodeoxygenation catalysts. Grand-canonical sampling of a sub-nanometer $Ru_{10}/TiO_2(101)$ cluster across $0 \leq n_{H2O} \leq 9$ on pristine and oxygen-vacancy supports shows hydroxyl returning as the dominant adsorbate-O species (45–50%), bound at single-Ru top sites on under-coordinated periphery atoms with no flat-metal analogue. An OH-rich 6-water phase is the predicted operational state across the 400–530 K aqueous-catalysis window. Hydrogen spillover to the support is an order of magnitude more probable on the reduced support at saturation coverage, but lies below the sampling resolution at the operational six-water plateau on both supports; Bader analysis shows the vacancy also changes its character, with the transferred hydrogen arriving hydridic on an under-coordinated Ti rather than protic on a lattice oxygen. The active site is the cluster periphery and the relevant adsorbate is OH, not the fully dissociated O + H of the flat-Ru paradigm.

1. Introduction

The extended single-crystal surface is the default reference for adsorbate speciation at metal surfaces, and for water on close-packed metals it is a well-characterized one. On Ru(0001) in ultra-high vacuum, partially dissociated bilayers of intact H_2O and OH are firmly established by both experiment and theory [1,2]; on warming, and in equilibrium with liquid water, ab initio thermodynamics instead predicts complete dissociation into atomic O and H above ~ 340 K, with no stable hydroxyl anywhere in the catalytic temperature range [3]. A working catalyst, however, is not an extended surface. Sub-nanometer metal particles dispersed on a reducible oxide expose a periphery of strongly under-coordinated atoms whose local d-band structure, and therefore whose adsorbate binding preferences, have no counterpart on a close-packed facet [4,5], and they sit on a support that can itself donate electrons (especially in the presence of O-vacancies) and accept protons. Whether flat-metal speciation survives that change of reference is the question addressed here.

The importance of this question is highlighted by the fact that ruthenium on titanium dioxide is one of the most effective catalyst families for the aqueous-phase upgrading of biomass-derived oxygenates [6]: optimized Ru/TiO_2 formulations achieve full xylose conversion at 99% xylitol selectivity on rutile TiO_2 [7] and efficiently hydrodeoxygenate real bio-oils and lignin-derived phenolics [8,9]. The experimental literature on these systems assigns water-derived species, surface hydroxyls and interfacial protons a mechanistic role in aqueous hydrogenation and hydrodeoxygenation [9,10], and operation is typically in the 390–500 K window. Flat-Ru thermodynamics places no stable OH anywhere in that window. The contradiction is unresolved, and it is the flat-metal reference, not the experiment, that we test here.

Over the past decade, experimental investigations of Ru/TiO_2 have established that catalytically active sites are primarily located at the metal–support interface, rather than on the metal surface itself. Omotoso and colleagues demonstrated that modifying the Ru loading, particle size, and TiO_2 phase attributes the direct C–O bond cleavage in m-cresol to perimeter sites; their DFT calculations showed a reduction in the energy barrier from 1.4 eV on the Ru surface to 0.7 eV at an interfacial site, and further to 0.3 eV when water donates a proton [9]. Operando DRIFTS and in-situ EPR studies have linked oxygen vacancies in Ru/TiO_2 to electron transfer from the support to interfacial Ru [11]. It has also been shown that, in the case of Ru supported on TiO_2, the metal–support interaction governs dispersion, activity, and selectivity in

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anisole hydrodeoxygenation [12]. In liquid-phase guaiacol conversion, hydrogen spillover from Ru to TiO_2 enhances activity, particularly on rutile compared to anatase [13,14]. On TiO_2, this spillover proceeds via coupled proton–electron transfer and is approximately ten orders of magnitude faster than on alumina [15].

These measurements do not reveal which water-derived species occupy the Ru surface under reaction conditions, and the DFT models used for interpretation [9,15,16] employ single static structures with arbitrarily chosen water content, rather than determining coverage by chemical potential. Recent theoretical studies have shown that this modeling choice significantly influences the results. Under reaction conditions, a supported sub-nanometer cluster comprises multiple thermally accessible isomers, with populations that depend on temperature and adsorbate chemical potential [17,18]. For graphite-supported Pt clusters, the active sites are determined by the applied potential [19], and for Pt nanoparticles in hydrogen, the most active sites are rare structural motifs rather than the most prevalent ones [20]. For Ru/TiO_2 in water, the three experimentally referenced quantities (the interfacial hydroxyl group, electron transfer from oxygen vacancies, and hydrogen transfer to the support) are all outcomes of an ensemble-based approach. None of these properties has yet been calculated for a supported Ru cluster with water coverage in equilibrium with the environment. Such a calculation would establish the adsorbate state necessary for interpreting the above observations and would directly test the flat-metal reference.

We treat the $Ru_{10}/TiO_2(1\,0\,1)$ interface as a fluxional cluster equilibrated with water at vapor saturation pressure rather than as a rigid surface, and find that this changes the predicted active state on three coupled axes: adsorbate speciation, phase composition and hydrogen flux. Grand-canonical global optimization produces a 1774-structure DFT dataset across $0 \leq n_{H2O} \leq 9$ on pristine and oxygen-vacancy supports. Hydroxyl is recovered as the dominant adsorbate-oxygen species on both supports, bound almost exclusively at single-Ru top sites on under-coordinated peripheral atoms; ab initio thermodynamics identifies an OH-rich phase of intermediate coverage ($n_{H2O} = 6$) as the operational state across the 400–530 K window; and hydrogen spillover onto the support, an order of magnitude more accessible on the vacancy at saturation, is absent from that operational plateau. Each of these results, together with the charge analysis associated with them, may be compared with a measurement mentioned above: namely, the top-site hydroxyl group together with the interfacial OH proposed for the Ru/TiO_2 hydrodeoxygenation process [9,10]; the lower charge of the dry cluster on the reduced support in connection with the vacancy-to-Ru electron transfer detected by EPR [11]; and the dependence of spillover on coverage, along with the hydride-like character of the hydrogen transferred on the reduced support, in light of the conflicting accounts of the role of water [21,22] and the coupled proton–electron mechanism of spillover on TiO_2 [15]. Section 2 sets out the computational protocol, Section 3 the results, and Section 4 what they imply for how supported sub-nanometer cluster catalysts should be modeled.

2. Computational methods

2.1. Density-functional set-up

All electronic-structure calculations were carried out with the Quickstep module of CP2K [23–25] using the Gaussian-and-plane-waves formalism. Valence states were expanded in a DZVP-MOLOPT-SR-GTH double-ζ basis [26] with a 400 Ry plane-wave density cut-off; core electrons were replaced by GTH pseudopotentials [27–29], retaining 1, 6, 12 and 16 valence electrons for H, O, Ti and Ru respectively. Exchange and correlation were treated with the PBE functional [30] augmented by Grimme's D3 dispersion correction with Becke–Johnson damping [31,32]. For structures containing an oxygen vacancy, an on-site Coulomb repulsion (U = 4.0 eV) was applied to the Ti 3d orbitals [33–36] to localize the excess electrons donated to the support by the vacancy [37,38]. SCF cycles were converged to 10^{-6} Ha at the Γ point.

2.2. Surface and cluster models

The supercell was a 4 × 2 two-trilayer slab of anatase $TiO_2(1\,0\,1)$ with cell parameters 18.98 × 20.33 × 20.00 Å; 20 Å of vacuum separated periodic images along z, and the bottom trilayer was held fixed at the bulk geometry. The oxygen-vacancy support was constructed by removing the two-coordinated bridging surface oxygen (O_{2c}) closest to the projected center of mass of the supported Ru cluster. Both the $Ru_{10}/TiO_2(1\,0\,1)$ model and the vacancy construction follow our earlier work on the same cluster–support interface [39], which validated the present grand-canonical approach on the dry cluster. A representative system, including the O-vacancy, is shown in Fig. 1.

The choices made for modeling this system are justified with regard to the literature and the questions being posed here. Dry Ru_{10} cluster morphology has recently been studied on the same support [39]. This cluster size is a good compromise: large enough to show genuine morphological diversity, small enough to sample that diversity successfully, and relevant to real-world catalytic systems. Anatase (1 0 1) is the majority facet of anatase nanoparticles and therefore the surface that a supported cluster predominantly presents. In reductive conditions, O-vacancies will appear in the support, but they are known to form immediately below the Ru clusters [39], rather than being isolated surface vacancies accessible to the solvent. The Coulomb correction on Ti 3d allows vacancy electrons to be localized correctly rather than smearing across the slab.

Because the object of the study is the state of water on the metal, water molecules were adsorbed on the Ru cluster only and the support beyond the cluster footprint was kept adsorbate-free. In liquid water the defect-free anatase (1 0 1) surface is predominantly molecularly hydrated [40–42]: the first layer is a regular array of intact molecules on Ti_{5c} sites, and conversion to bridging hydroxyl requires either a surface defect [43] or an excess electron delivered at the aqueous interface [44]. This near-constant solvation background cancels in the relative free energies that set the phase boundaries: across the low-energy ensemble the van der Waals footprint of the cluster is 59 ± 4 $Å^2$, independent of water coverage and of support (Section S9). This approximation is common in grand-canonical global optimization of supported clusters and has been previously applied for the study of hydrogen coverage [45–47]. It is extended here to the study of water coverage.

2.3. Grand-canonical configurational sampling

Structures were generated in two phases (Fig. 2). In Phase 1, the PGOPT software [39,48] generated 10,000 candidate structures, each combining a randomly generated Ru_{10} cluster with 0–8 adsorbed water molecules and an explicit enumeration of dissociation states, that is, of the number of broken O–H bonds; combinations of five or more waters with eight or more broken bonds were excluded. Each candidate was optimized in two stages, a pre-optimization with an SVP basis on Ti followed by full re-optimization at the production DZVP-MOLOPT level. Geometric filtering then removed structures with H atoms residing inside the TiO_2 support, and structures in which any adsorbate O was not bonded to the Ru cluster (with an Ru–O distance cut-off of 2.29 Å, 120% of the sum of the covalent radii), leaving 1377 Phase-1 structures.

Phase 2 targeted the under-sampled cells of the (n_{H2O}, n_{broken}) sampling matrix carrying five to eight waters and zero to five broken bonds. For each such cell the Phase-1 minimum was used as a seed and one further water molecule was placed at a random position around the cluster using the random-placement utilities of GOCIA [49], with 50 trials per dissociation state of the added water; empty cells were initialized from the corresponding (n − 1) seed. The resulting 3377 candidates were pre-optimized over ten geometry steps, geometrically filtered and ranked, and the twenty lowest-energy structures within 1 eV of each per-cell minimum were fully relaxed at the production level.

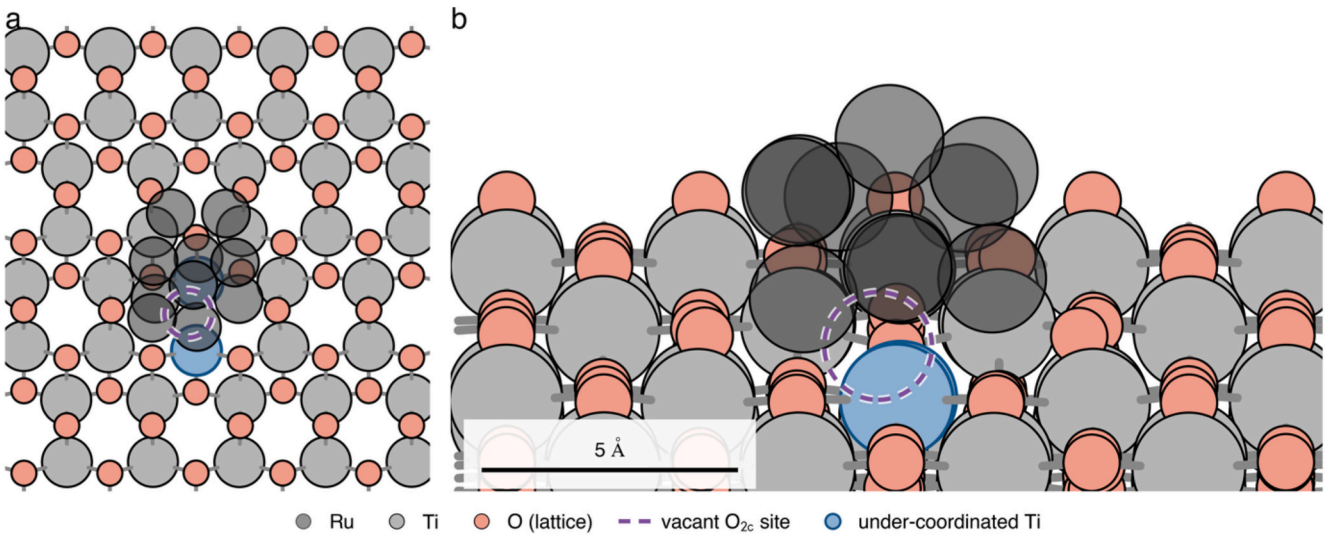


Fig. 1. Model of the reduced $Ru_{10}/TiO_2(101)$ interface, shown for the dry vacancy global minimum. (a) Top view. The dashed circle marks the vacant bridging surface oxygen (O_{2c}) site; the two Ti atoms left under-coordinated by its removal are shaded. (b) Side view of the same structure.

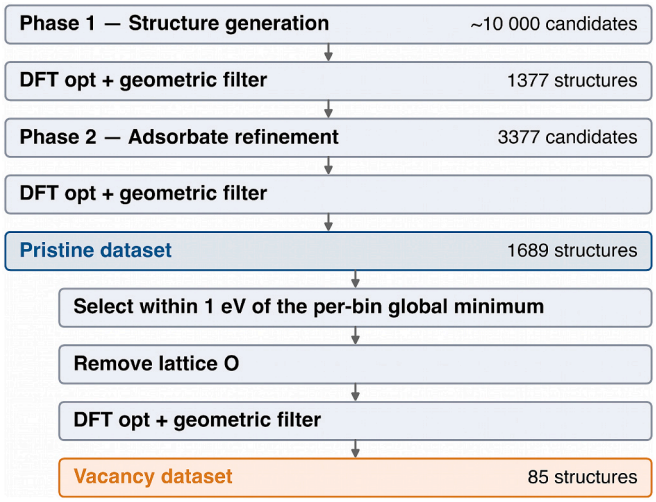


Fig. 2. Two-stage workflow for sampling the coverage and conformations of supported Ru clusters and generating two databases of structures. Full workflow is detailed in Fig. S1.

The vacancy dataset was built by selecting, from the pristine dataset, every structure for which there exists a temperature between 200 and 700 K at which it lies within 1 eV of the global minimum, removing the lattice oxygen closest to the cluster center of mass, and re-relaxing at full DFT. This avoids repeating the entire global optimization, a shortcut we have previously shown to work [39]. The final dataset contains 1689 pristine and 85 vacancy structures. Sampling-sufficiency diagnostics are given in Section S2.

2.4. Adsorbate classification and ensemble averaging

The sampling procedure stores, for each candidate, an integer water count together with bookkeeping of the dissociation moves applied during the search. Two factors make those stored labels unsafe downstream: on a fluxional cluster, late-stage relaxation can move H atoms between adsorbates without flipping the bookkeeping flags, and water molecules that desorb from the cluster are removed. Every adsorbate is therefore reclassified a posteriori from the post-relaxation atomic positions alone, using bond cut-offs of 1.2 × the sum of the covalent radii. Adsorbate-derived oxygens are classified as intact H_2O (two bonded H), hydroxyl (one bonded H) or bare O (no bonded H); adsorbate-derived hydrogens as Ru-bound (bonded to a cluster Ru atom and to its parent O), spillover (bonded to a substrate Ti or substrate O at the cluster–support interface), or part of an intact H_2O or OH. The broken-bond descriptor is recomputed as $b = n_{OH} + 2n_{O,bare}$. Three sanity checks are imposed on every structure: total adsorbate O equal to n_{H2O}, total adsorbate H equal to $2n_{H2O}$, and every adsorbate O bonded to between one and four Ru atoms. All 1774 structures pass. Because a hydrogen at the cluster–support contact can lie within bonding distance of both a cluster Ru and a substrate Ti, the assignment follows a fixed order: a hydrogen within 1.20 Å of an adsorbate O belongs to that hydroxyl or water molecule; failing that, a hydrogen within 2.00 Å of both a cluster Ru and a substrate Ti is an interfacial bridge; failing that, a hydrogen within 1.20 Å of a lattice O, or within 2.00 Å of a substrate Ti with no Ru inside 2.00 Å, has spilled onto the support; the remainder are Ru-bound. We report the bridge class separately in Section 3.6, where we describe its consequences for the spillover statistics.

All reported species fractions, binding-mode distributions, coordination distributions and spillover probabilities are Boltzmann-weighted ensemble averages, $p_i \propto \exp(-G_i/k_BT)$, normalized within each (support, n_{H2O}) bin and evaluated at $T = 500$ K unless stated otherwise. They are not raw counts over the sampled structures, so a bin does not acquire weight in a given species merely by holding more structures.

2.5. Ab initio thermodynamics

Ab initio atomistic thermodynamics [50,51] converts each structure's DFT total energy into a Gibbs free energy of adsorption. For a structure containing n_{H2O} water-derived oxygen atoms,

$$\Delta G(T,p) = E_{nH2O} - E_0 - n_{H2O}\mu_{H2O}(T,p) \quad (1)$$

referenced to the dry-cluster global minimum E_0 on the same support and to liquid water at saturation pressure. Phase diagrams are constructed by evaluating Eq. (1) for the global minimum of each (support, n_{H2O}) block and reading off the lower envelope of the resulting family of $\Delta G_{min}(T)$ curves.

2.6. Bader charge analysis

Bader charges were computed by single-point DFT at the relaxed geometry of the eight phase-diagram global minima (pristine $n_{H2O} = 0$, 2, 6, 9 and vacancy $n_{H2O} = 0$, 1, 6, 9), using a grid-based decomposition

of the all-electron density [52–55].

2.7. Scope of the thermodynamic treatment

The approach described above is purely thermodynamic, which requires an assumption on the relative kinetics of adsorbate equilibration and catalytic activity. The adsorbate layer equilibrates quickly compared with catalytic turnover. Barriers for water dissociation on ruthenium are modest: about 0.80 eV for a chemisorbed monomer on close-packed Ru(0001) [1], falling to 0.69 eV on Ru_8 and 0.59 eV on Ru_{14} [56]. A NEB calculation performed on one of the Ru_{10} clusters in the dataset yielded a barrier of 0.67 eV. Furthermore, on Ru(0001) the barrier has been shown to drop to 0.59 eV in the presence of a second water molecule, an effect which is also expected on Ru clusters [57]. With a harmonic prefactor of 10^{13} s^{-1}, 0.80 eV corresponds to 1 ms at 400 K and 10 μs at 500 K; over the seconds-to-hours lifetime of a working catalyst at 400–530 K, the layer is therefore equilibrated, and the populations reported here govern speciation.

2.8. Calculation of solvation free energies

All previously reported results pertain to the solid–gas interface and leave out the possibility of water molecules being adsorbed on the support. A direct investigation was conducted to determine whether contact with liquid water alters the phase diagram.

The solvation free energies of the various cluster/water configurations obtained above were calculated using SolvHybrid, a method based on the hybrid quantum–mechanical/molecular-mechanical approach developed by Clabaut et al. [58]. Periodic DFT is used to describe the electrostatics of the interface and the adsorbate/support energy, and MM calculations then calculate, by means of alchemical thermodynamic integration, the contribution of solvation to the adsorption free energy. As well as bridging the gap between simulating the surface state at the solid–gas interface at water vapor saturation pressure, and the solid–liquid interface at that same chemical potential, this method addresses the unresolved issue of water coverage on the support beyond the cluster. The sampling described above focused on adsorbing water molecules on the Ru_{10} cluster; the solvation technique employed here solvates the entire slab, as shown in Fig. S8. Full details are given in Section S11.

3. Results and discussion

3.1. Adsorbate speciation: hydroxyl dominates on both supports

The grand-canonical sampling on Ru_{10} produces 1689 structures on the pristine support and a more highly selected 85 on the vacancy support. The 85 vacancy structures are selected from the most stable structures without a vacancy so as to avoid needing to repeat the entire global optimization, a method we have previously shown to work [39]. We classify every adsorbate-derived oxygen as intact H_2O, OH or bare O on the basis of the number of bonded protons (cut-off $d_{O-H} < 1.20$ Å) and of its first-shell coordination to Ru (cut-off $d_{Ru-O} < 2.60$ Å). Reported species fractions are Boltzmann-weighted ensemble averages at 500 K (weights normalized within each n_{H2O} bin; Section 2.4), not raw counts over the sampled structures. On both supports, partial dissociation is the dominant outcome: OH accounts for 44.9% of all adsorbate oxygens on the pristine support and 50.2% on the vacancy support, against 54.0%/45.4% molecularly intact and 0.6%/3.8% fully dissociated bare-O.

The canonical flat-Ru reference state at the same operating temperature, explicitly equilibrated with liquid water and surveyed by recent ab initio thermodynamic studies [3], predicts complete dissociation of adsorbed water into atomic O and H above ~ 340 K, with no stable hydroxyl species at catalytic temperatures. On flat Ru(0001) at low temperature in UHV, partially dissociated bilayers of OH and intact H_2O are well documented [1,2], but those mixed phases destabilize on warming and survive neither on the bulk metal in aqueous conditions nor at catalytic temperatures. The ground-state ensemble we sample here on the Ru_{10} cluster recovers OH as the dominant adsorbate-O species under aqueous conditions at 300–530 K, lifting the flat-metal/working-catalyst contradiction noted in the introduction.

3.2. Adsorbate binding mode

Fig. 3a,b complements the speciation picture with the binding-mode statistics (likewise Boltzmann-weighted at 500 K; Section 2.4), providing the distribution of OH (and of intact H_2O and bare O for context) over the number of Ru atoms each adsorbate is bonded to (1 = top, 2 = bridge, 3 = hollow), as well as the Ru–Ru coordination distribution. On the pristine support, 89% of OH groups bind in the *top* mode: they coordinate to exactly one Ru atom of the cluster. The vacancy figure is 82% on top sites; the remaining ~ 18% is split between two-coordinated bridge sites and a small contribution from threefold hollow sites (Fig. 3b). Intact H_2O is even more top-selective (essentially 100% in top mode on both supports), while bare O distinctively populates a high-coordination (4-Ru) basin on the pristine support but disappears on the vacancy.

We next test whether the preference for top-mode binding is driven by the under-coordination of the host Ru atom rather than by sterics alone. To check this, we compute the Ru–Ru cluster coordination of every Ru atom in every relaxed structure (number of intra-cluster Ru neighbors within 3.0 Å) and compare two distributions (Fig. 3c,d): all Ru atoms in the Boltzmann ensemble at 500 K (open bars), and the subset that hosts at least one adsorbate (filled bars). Bulk Ru is 12-coordinated; on the 10-atom cluster the coordination number ranges from 1 to 7, with a peak at 3–4 on both supports. Conditioning on adsorbate occupancy shifts the distribution to lower coordination: on the pristine support the fraction of two-coordinated Ru atoms nearly doubles (9.4% all vs. 15.5% adsorbate-bearing), and the mean coordination of host atoms drops from 3.86 to 3.51. The vacancy support shows a smaller but consistent shift (3.92 to 3.75). Adsorbates therefore decorate under-coordinated peripheral Ru atoms preferentially, in line with the Calle-Vallejo coordination-number framework [4,5]. Under-coordinated metal atoms experience an upshift of the local d-band and bind adsorbates more strongly; σ-bonded top-site geometries are thereby favored over the multi-center hollow geometries that flat, fully coordinated surfaces stabilize.

If hydroxyl is the catalytically relevant proton-transfer species in aqueous hydrogenation [9], then it resides at the periphery of the cluster, not on flat terraces, where in liquid–water equilibrium it does not exist as a stable species in the first place [3]. The flat-Ru reference state, fully dissociated water (atomic O + 2H, both O–H bonds broken) on the bare metal terrace, is not the right model for the active site of a working Ru/TiO_2 catalyst.

3.3. Ab initio thermodynamic phase diagrams

We convert the static structural ensembles into thermodynamic phase information by applying Eq. (1) to the global minimum of each (support, n_{H2O}) block to obtain $\Delta G_{min}(T)$ per coverage. Plotting all coverage curves on a single ΔG vs. T axis and reading off the lower envelope (Fig. 4) yields the ab initio thermodynamic phase diagram.

On the pristine support (Fig. 4a), the system saturates at $n_{H2O} = 9$ at 300 K (ΔG_{min}, the Gibbs free energy of the most stable structure at that coverage, = −2.77 eV referenced to the dry cluster) and crosses to the low-coverage 2-water phase by 365 K, with a formally stable 6-water window of only ~ 6 K (359–365 K) in between. We read this sliver as a near-degeneracy of the 9-, 6- and 2-water curves around 360 K rather than a thermodynamically resolved phase. The 6-water composition is not operationally significant on the pristine support.

Fig. 4b shows the phase diagram for the oxygen-vacancy slab. The

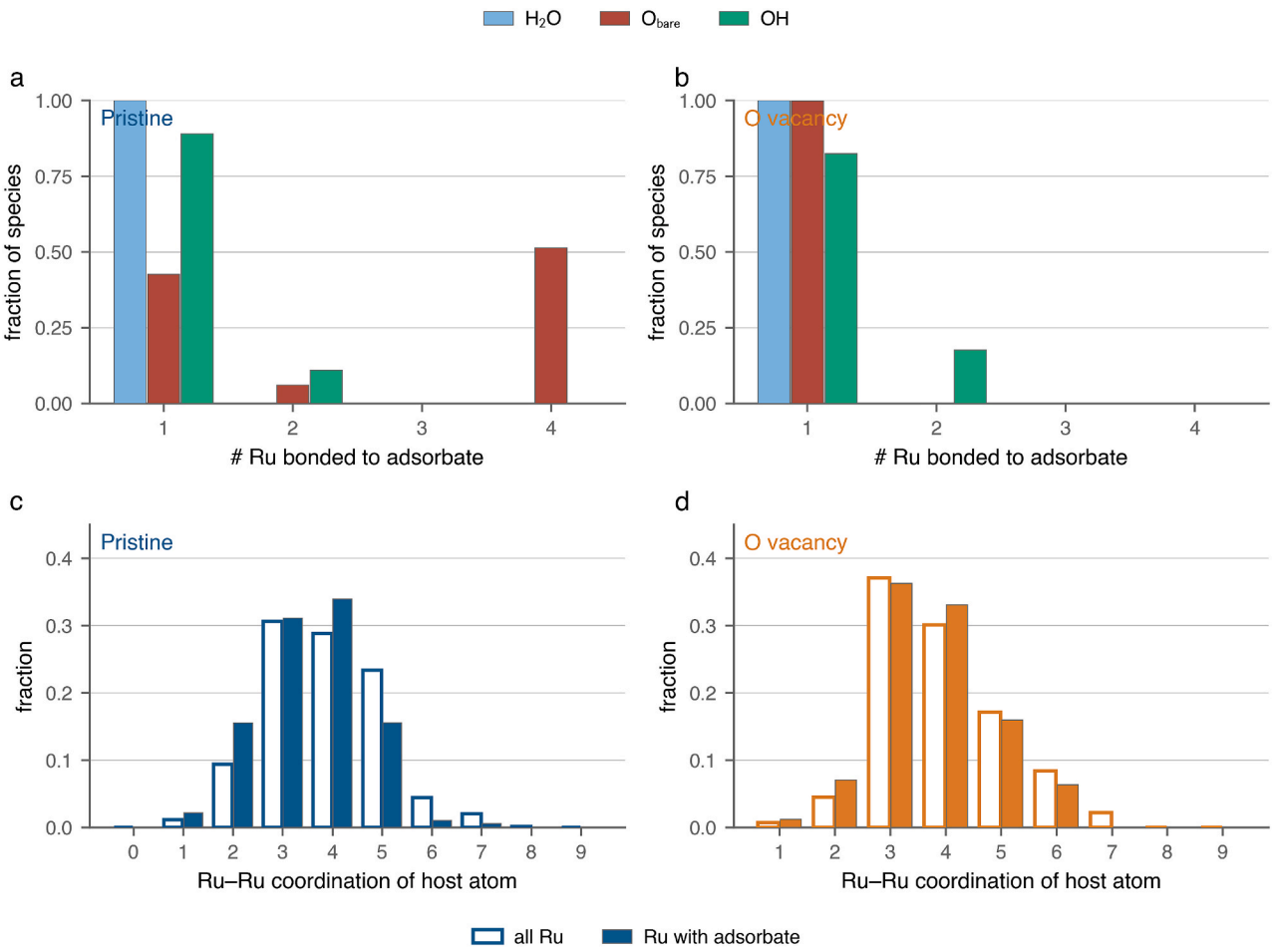


Fig. 3. (a, b) Distribution of three adsorbate species (intact H_2O, OH and bare O) over the binding mode (number of Ru atoms each adsorbate is bonded to: 1 = top, 2 = bridge, 3 = hollow), pooled across all relaxed structures on each support and Boltzmann-weighted within each (support, n_{H2O}) bin at T = 500 K and water saturation pressure (catalytic reference, see Section 2.4). (a) Pristine TiO_2: 89% of OH on top sites. (b) Vacancy TiO_2: 82% on top sites. (c, d) Distribution of the Ru–Ru coordination number (number of neighboring Ru atoms) of cluster Ru atoms, computed by enumerating every Ru atom in every relaxed structure within the same Boltzmann-weighted ensemble: open bars give the distribution over all cluster Ru atoms; filled bars restrict to the subset that hosts at least one adsorbate. Mean Ru–Ru coordination: pristine 3.86 (all) → 3.51 (adsorbate-bearing); vacancy 3.92 (all) → 3.75.

saturated 9-water phase remains the 300 K minimum, with a markedly exergonic $\Delta G_{min} = -12.4$ eV referenced to the unhydrated vacancy cluster. The unusually large 300 K stabilization reflects the fact that the unhydrated vacancy cluster is itself a high-energy reference (the Ti^{3+} polaron and the dangling under-coordinated Ru sites are both healed once water is admitted); the chemical potential implied by extrapolating the saturated branch back to liquid–water equilibrium sits well above $\Delta\mu^{liq}_{H2O} \approx 0$ eV at 300 K (Section S3), confirming that the $n_{H2O} = 9$ saturation cap of the present search is a lower bound on the cluster's water loading at room temperature rather than the equilibrium loading. The 6-water plateau identified below is not affected by this cap: it lies well inside the sampled range and is bracketed by lower- and higher-coverage curves that the search resolves explicitly. The absolute energetics of the 300 K saturation point should therefore not be over-interpreted. The 9-water curve depends steeply on T and crosses the 6-water envelope at 366 K. Between 366 and 529 K the most stable composition is $n_{H2O} = 6$, a true plateau spanning the operationally relevant 400–530 K aqueous-catalysis window. Above 529 K the envelope falls again, this time to the one-water curve. The vacancy therefore introduces a thermodynamic plateau between the saturated and low-coverage regimes accessible on the pristine support, and we assign this intermediate-coverage configuration as the operational state of the catalyst.

3.4. The six-water plateau

The chemistry behind the new plateau follows from the speciation analysis above. On the vacancy global minimum for $n_{H2O} = 6$, three of the six adsorbate oxygens are present as OH, two as intact H_2O, and one as a fully dissociated bare O on the cluster periphery (Fig. 5, second column from the right; Vacancy $n_{H2O} = 6$ panel). The vacancy site itself is not healed by an O atom (geometric inspection of the 6-water minimum places the bare O ~ 5.6 Å from the vacancy), so the role of the defect is not to be sterically healed by an adsorbate but to donate excess electrons to the metal phase, lowering the effective work function of the cluster periphery and stabilizing the OH-rich shell that defines the plateau.

Operando hydrogenation and HDO on Ru/TiO_2 typically run in the ~ 390–500 K window (xylose hydrogenation, phenolics-HDO and bio-oil upgrading; see, e.g., refs [7–9]); the upper end of this window sits squarely inside the vacancy 6-water plateau identified above, while the lower end overlaps with the high-temperature edge of the saturated 9-water phase. On a pristine flat support, a static surface model would put the catalyst in its low-coverage two-water state; on a defective support our calculations identify a thermodynamically robust, OH-rich 6-water phase as the operative state. The two predictions imply different active sites and different proton-shuttling networks. The reported support-dependence of Ru/TiO_2 selectivity [7] is consistent with

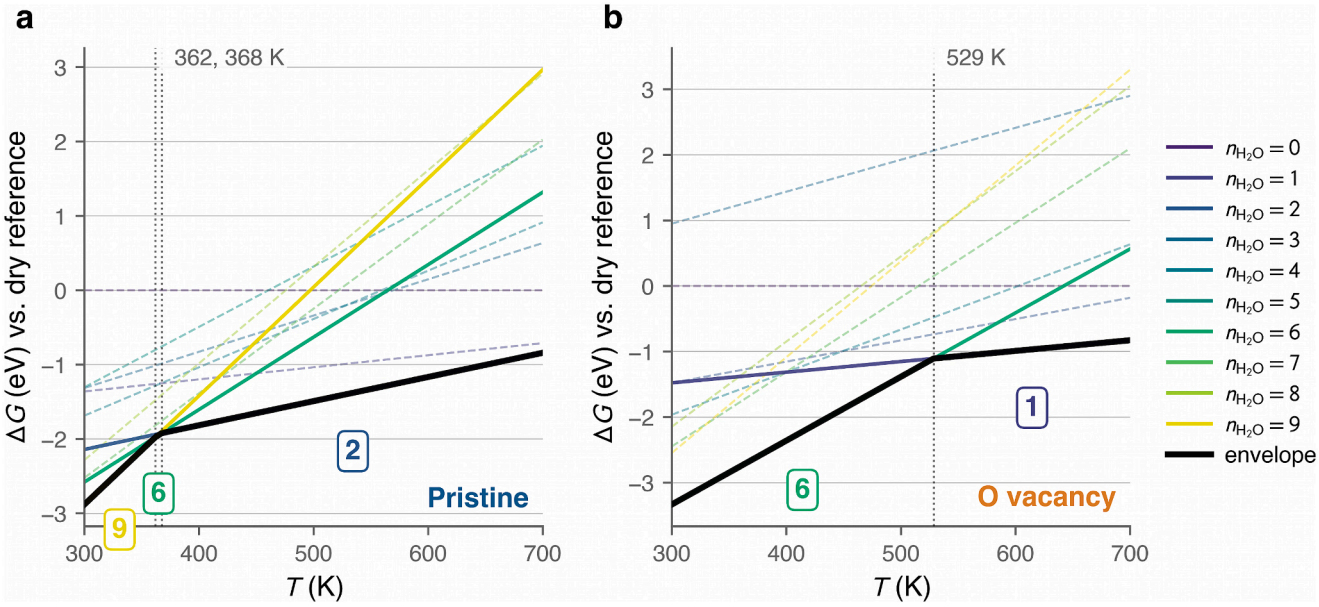


Fig. 4. *Ab initio* thermodynamic phase diagrams of the $Ru_{10}/TiO_2(1\,0\,1)$ interface at equilibrium with water at vapor saturation pressure, in the canonical $\Delta G(T)$ representation. Each thin curve is the Gibbs free energy of the bin global minimum, $\Delta G_{min}(T)$ (Eq. (1)), for one water coverage n_{H2O} (curves colored by water loading), referenced to the dry-cluster minimum on the same support; coverages contributing to the lower envelope are drawn solid, the others dashed. The black line is the lower envelope, the most stable composition at each T. Vertical dotted lines and accompanying labels mark the phase-transition temperatures (where two adjacent coverages cross); the boxed numerals identify the value of n_{H2O} of the most-stable composition that defines each plateau. **(a)** The pristine support passes from the saturated 9-water phase through a narrow ~ 6 K window of formally stable 6-water (between 359 and 365 K) to the low-coverage 2-water phase. **(b)** The vacancy support features an OH-rich 6-water plateau up to 529 K, spanning the operationally relevant aqueous-catalysis window. A 2D-heat-map view of the same data is given in Fig. S5.

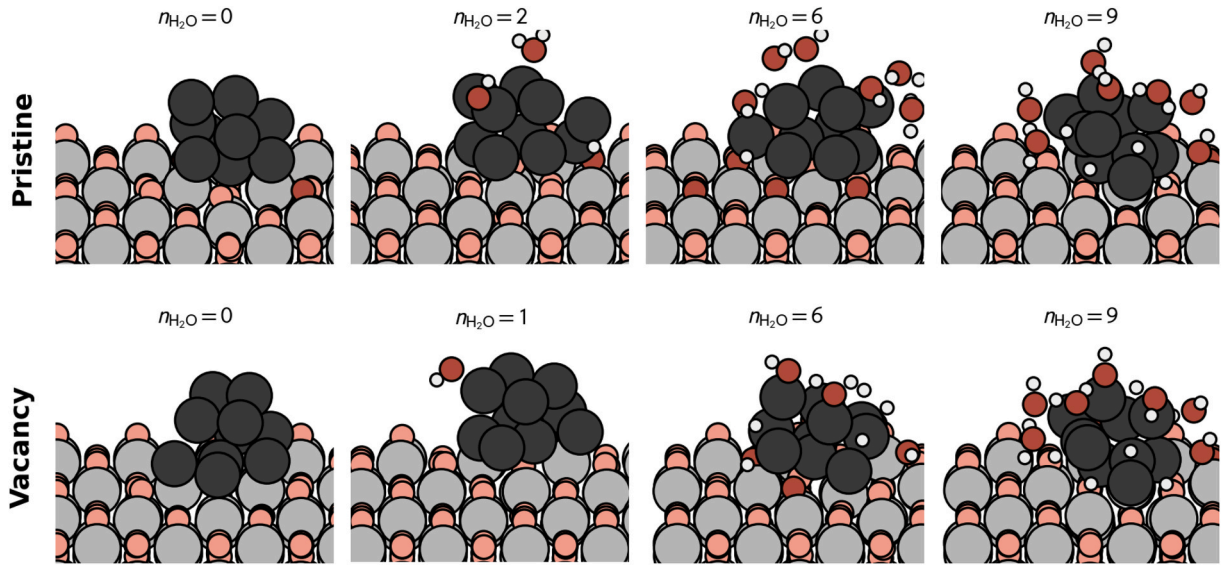


Fig. 5. Side-on views of the eight phase-diagram global minima covering the temperature range, arranged by support: top row, pristine (n_{H2O} = 0, 2, 6, 9); bottom row, vacancy (n_{H2O} = 0, 1, 6, 9). Each panel is a ≈13 Å window centered on the cluster, retaining only the top rows of the TiO_2 slab. Charcoal: Ru; light gray: Ti; salmon: lattice O; brick red: adsorbate O; near-white: H. Top-down views of the full slabs are given in Fig. S6.

the latter picture.

3.5. Cluster morphology and electronic response

We characterize each cluster geometry by three Boltzmann-weighted metrics at 500 K: the asphericity $\delta = (I_3 - I_1)/(I_3 + I_1)$ of the gyration tensor, where $I_1 \leq I_2 \leq I_3$ are the principal moments; the cluster height Δz between the highest and lowest Ru atoms; and the center-of-mass height $z_{COM,Ru}$ above the substrate (Fig. 6a–c). The response is non-monotonic on both supports. The cluster passes through an oblate, low-δ state at low coverage, an elongated peak near $n_{H2O} = 4$ ($\delta = 0.50$ pristine, 0.57 vacancy), and back to a more spheroidal arrangement at saturation. Cluster height tracks asphericity, the most spheroidal configurations also standing tallest, and the vacancy systematically pulls $z_{COM,Ru}$ down relative to pristine, consistent with stronger interfacial adhesion when an O vacancy sits beneath the cluster. The two supports follow one another closely except at the catalytically relevant $n_{H2O} = 6$ plateau, where the vacancy cluster is markedly more spheroidal

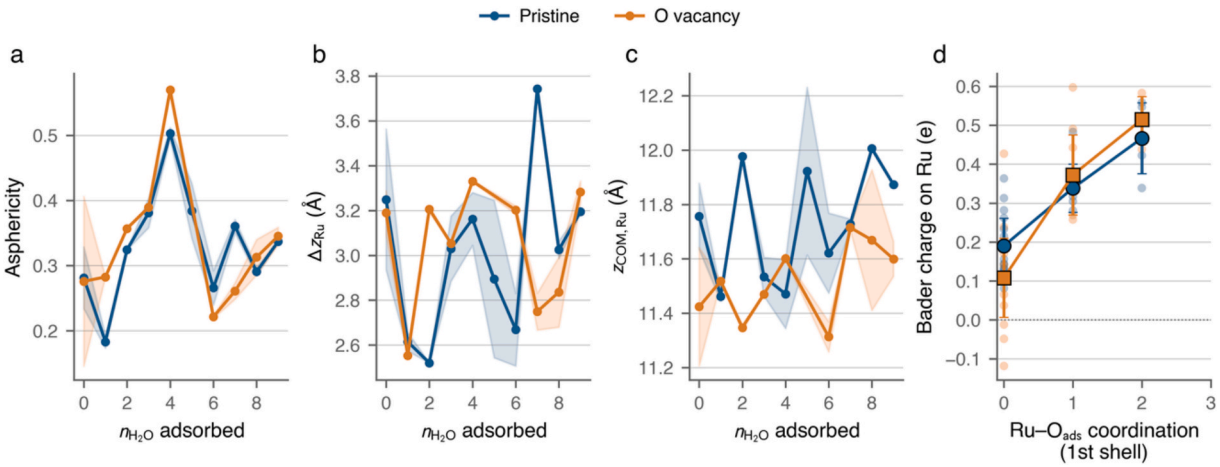


Fig. 6. Cluster morphology and electronic state at $T = 500$ K. (a) Asphericity δ against water coverage. (b) Cluster height Δz. (c) Center-of-mass height of the Ru cluster above the substrate. Curves in (a)–(c) are Boltzmann-weighted ensemble averages within each (support, n_{H2O}) bin; shaded bands give the $\pm 1\sigma$ structure-to-structure dispersion. (d) Per-Ru Bader charge as a function of the number of adsorbate O atoms in the first coordination shell, evaluated on the eight phase-diagram global minima (small dots: individual Ru atoms; large markers: weighted means).

($\delta = 0.22$) than its pristine counterpart ($\delta = 0.26$).

Bader analysis of the eight phase-diagram global minima resolves the electronic restructuring behind that morphological response (Fig. 6d). The per-Ru charge increases monotonically with the local Ru–O_{ads} first-shell coordination, from $\langle q_{Ru} \rangle = +0.13$ e at zero adsorbate contact to + 0.50 e at two, at roughly + 0.18 e per contact and with the two supports overlapping to within statistical noise. Integrating over all ten Ru atoms, the total cluster charge rises from + 1.6 e (pristine, dry) to + 3.9 e (pristine, nine waters), while the dry cluster on the vacancy starts 0.9 e lower at + 0.7 e. We read this as partial transfer of the vacancy-liberated polaronic electrons from Ti^{3+} centers into the Ru cluster, which acts as a soft electron sink for charge that would otherwise sit on the bare oxide. As water is added the donor effect is progressively overwhelmed by cluster–adsorbate charge transfer, and the two supports converge to within ~ 0.7 e at saturation (Table S2).

Adsorbate OH oxygens carry − 1.1 to − 1.5 e, more negative at higher coverage, and the bonded H carries + 0.9 e. The net ~ − 0.5 e per OH unit is intermediate between molecular hydroxide and a covalent surface OH ligand, consistent with a partially anionic σ-bonded hydroxyl on a Lewis-acidic Ru top atom: the Ru–O bond carries substantial covalent character but the oxygen retains a clear electron surplus. The pronounced O–H polarization, an anionic oxygen against a markedly protic hydrogen, marks this hydroxyl as Brønsted-acidic, poised to donate its proton to the support or the solvent rather than to abstract one. That is the pathway examined in Section 3.6. Full per-species Bader charges are tabulated in Section S6.

3.6. Hydrogen spillover

We also observed spontaneous hydrogen spillover (the migration of a hydrogen atom from the metal to the support) in a substantial fraction of our sampled structures (~18%). Since hydrogen spillover has been invoked as a key step in Ru/TiO_2-catalyzed hydrogenation [13] and is known to be sensitive to both support composition and the local water environment of the cluster (loading and degree of H-bonding) [14,15,21,59], we analyzed in more detail the factors influencing the efficiency of the spillover.

To quantify how often, and under what conditions, spillover configurations are sampled, our dataset distinguishes H atoms attached to Ru ("Ru-bound") from those attached to a substrate Ti or substrate-O atom ("spillover"). Hydrogen spillover is then evaluated using the Boltzmann-weighted probability $p(\geq 1$ spillover H) at the saturation

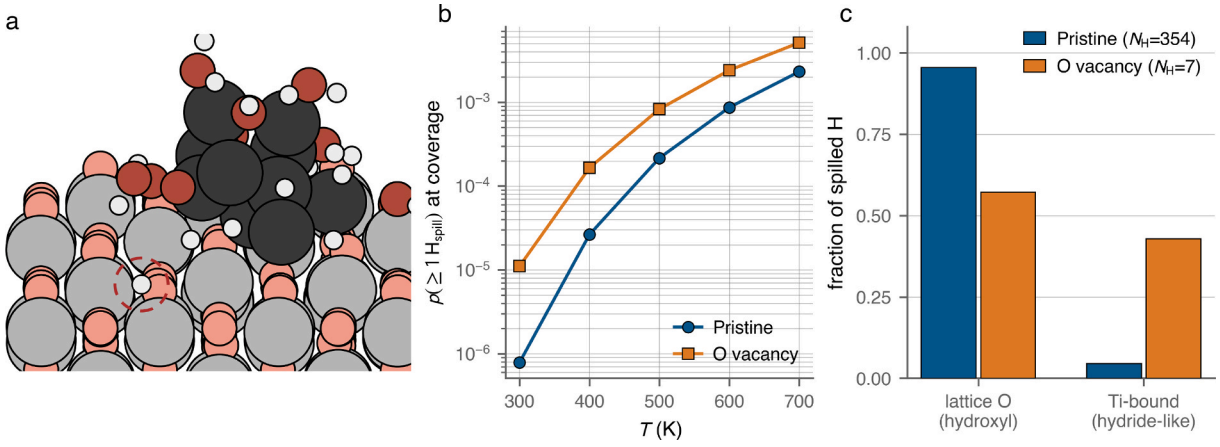


Fig. 7. Hydrogen spillover. **(a)** Representative spillover configuration: vacancy support at saturation ($n_{H2O} = 9$), side-on view. The migrated H atom (small white sphere, dashed ring) sits on a substrate Ti at the cluster–support interface, distinct from the cluster-bound OH and H_2O above. **(b)** Boltzmann-weighted spillover probability at the 300 K-most-stable composition ($n_{H2O} = 9$ on both supports), plotted against temperature on a log scale. The vacancy is ≈14 × more spillover-permissive than pristine at 300 K, with the advantage decaying to ≈2 × at 700 K as both probabilities approach unity. (c) Fraction of spilled hydrogen bound to support O atoms and support Ti atoms in the dataset, with and without an O-vacancy (N_H indicates the number of H atoms spilled over in the entire dataset).

coverage $n_{H2O} = 9$, which is the 300 K minimum on both supports (Fig. 7b). At 300 K the vacancy support is roughly an order of magnitude more spillover-permissive than pristine ($p_{vac} \approx 1 \times 10^{-5}$, $p_{prist} \approx 8 \times 10^{-7}$, ratio $\sim 14 \times$); both probabilities rise sharply with temperature, and the vacancy advantage decays to $\sim 6 \times$ at 400 K, $\sim 4 \times$ at 500 K and $\sim 2 \times$ at 700 K, by which point both probabilities are at the few-percent level. The 300 K ratio is built from only seven spillover-bearing structures across the entire vacancy ensemble (Section 2.4): the Poisson uncertainty on the count alone gives a multiplicative uncertainty of order $\pm$ 40% on the vacancy probability and hence on the ratio. Spillover is roughly an order of magnitude more accessible on the vacancy at saturation. The precise factor of fourteen should not be over-interpreted.

Across the 321 spillover-bearing structures, 361 such hydrogens appear in two classes (Section S10). On the pristine support, 338 of 354 sit on a lattice oxygen (in close proximity to the cluster; the dataset does not sample migration away from the cluster, beyond the distances available within the periodic slab used here) as an ordinary hydroxyl, with d(O–H) = 0.97–1.13 Å (median 0.99 Å); the other 16 sit at a bridge between a cluster Ru and a substrate Ti. On the vacancy support, the two classes occur in a 4:3 ratio (Fig. 7c).

While the hydrogen atoms adsorbed on support O atoms feature a positive charge, the bridging H atoms are hydride-like bearing a negative charge. It is of particular note that, on the support with an O-vacancy, the Ti atom the H atoms are bound to is consistently the first neighbor of the missing O atom. Single-point Bader analysis of these configurations (Section 2.6) gives the transferred hydrogen on the vacancy support q = − 0.27 e, while the hydroxyl hydrogens on the pristine support carry + 0.71 and + 0.77 e. Thus, the O-vacancy favors hydrogen spillover to the support by enabling hydrogen to transfer charge away from the support itself. These hydridic H atoms are also of interest when considering catalytic reactivity.

The spillover effects occur at high water coverage ($n_{H2O} = 9$), not at the catalytically relevant $n_{H2O} = 6$ plateau identified in the phase-diagram analysis above. This is a genuine coverage effect rather than an artifact of how each coverage was sampled: $p(\geq 1$ spillover H) is Boltzmann-weighted and normalized within each n_{H2O} bin (Section 2.4), so a bin does not appear more spillover-prone merely by holding more structures, only the energies of its spillover configurations enter. Spillover emerges at $n_{H2O} = 9$ because only at high water loading is the adsorbed layer connected enough to carry a hydrogen from the cluster onto the support; the sparser 6-water layer cannot. At the 6-water plateau itself $p(\geq 1$ spillover H) is below the resolution of the present sampling on both supports: no spillover-bearing structures sit within $3k_BT$ of the bin global minimum (although 41 of the 162 sampled structures at this coverage carry at least one spillover H, they all lie well outside that energy window), and a naive Boltzmann extrapolation places the probability several orders of magnitude below 10^{-6} at 300 K, rising only to the percent level by 700 K. The numerical extrapolated value is not chemically meaningful below the floor of our sampling; the substantive point is that spillover is essentially absent from the operational 6-water plateau on either support. These spillover probabilities are evaluated for the water-derived hydrogen of our reservoir, with no explicit H_2; under operando H_2 pressure, which supplies additional surface hydrogen, they should be regarded as lower bounds. Spillover is therefore not a feature of the operational steady state; instead, the vacancy opens a kinetically accessible spillover channel within the high-coverage regime, under the conditions encountered during transient flooding (start-up, rinse cycles, condensation events at the cluster periphery) rather than during isothermal turnover at the 6-water plateau.

What is described here captures the initial steps of hydrogen spillover from the Ru cluster onto the TiO_2 support.

3.7. Implications for the cluster–support interface

Mechanistically, the coverage dependence is consistent with the water-assisted spillover picture established for Pt/MOF hosts [21]: only the dense water and OH bath at $n_{H2O} = 9$ makes the transfer accessible at all, and on the reduced support the defect stabilizes the transferred hydrogen by binding it beside an under-coordinated Ti. The defect is therefore better described as the kinetic gateway for water-assisted spillover under transient flooding rather than as a permanent feature of the operational plateau. This gives a testable molecular handle for the observed support dependence of Ru/TiO_2 hydrodeoxygenation selectivity [7,9]: the active state in steady operation is the OH-rich, low-spillover 6-water plateau, but transient excursions to higher coverage open a defect-mediated spillover channel that is essentially unavailable on the pristine support.

Read together, the three results describe a cluster–support interface in which neither partner is a spectator. The metal supplies under-coordinated peripheral atoms whose local geometry selects σ-bonded top-site hydroxyl over the multi-center hollow adsorption that a close-packed terrace imposes; the oxide supplies electrons that stabilize the OH-rich shell, and accepts protons through the interfacial water network once the coverage is high enough to connect them. Neither channel is available to a single-crystal model, and between them they change which species is dominant, which composition is stable at operating temperature, and whether hydrogen can leave the metal.

How much of this phenomenon extends to other cluster sizes? The mechanism is expected to apply. The selection of the top-site hydroxyl is determined by the low Ru–Ru coordination among the peripheral atoms (Section 3.2), and any sub-nanometer cluster exposes atoms with coordination numbers of five or below. Consequently, the preference for a σ-bonded hydroxyl over the multi-center hollow adsorption is a property inherent to small Ru clusters, not just Ru_{10}. The vacancy effect is predominantly electronic, and its magnitude is expected to scale inversely with the number of cluster atoms sharing the electrons donated by the defect: smaller clusters should experience this effect more profoundly, whereas larger clusters should experience it to a lesser extent. Furthermore, the phase diagrams of similar-sized clusters (approximately 5–20 Ru atoms) should be expected to behave in similar ways, with the possible exception of magic numbers (Ru_8, Ru_{12}, Ru_{18}).

3.8. From the solid–gas to the solid–liquid interface

Using the solvation method described above (and more fully in Section S11), the solvation free energies of the global minima were calculated. This results in the phase diagram of the Ru/TiO_2 system at the solid–liquid interface (Fig. 8), which shows that solvation does not significantly impact the phase transitions. This in turn validates the sampling approximation made above when water was only explicitly sampled on the Ru cluster and not on the support, as the solvation calculation fully solvates the support, leading to no substantial change.

The most significant difference is the disappearance of the 6-water phase on the pristine support (which only had a small 6 K stability window before solvation). The other changes are minor shifts in phase-transition temperatures. The direction of these shifts is not a simple displacement in one direction and is not reducible to a simple explanation. This matches what is already known for the analogous situation on Ru surfaces rather than clusters [3].

A further possible effect of solvation was investigated: whether the presence of adsorbed water molecules on the support could provide a potential for hydrogen mobility away from the cluster through an H-bond network. However, this possible mechanism appears thermodynamically prohibited do to the cost of forming H_3O^+ by transferring a proton away from the Ru cluster (+0.83 eV in the case of an Ru atom bound directly to Ru, +0.97 eV in the case of an H atom from Ru-OH, and + 1.02 eV in the case of an H atom from Ru-OH_2).

Overall, solvation has a minimal impact on the system, changing phase-transition temperatures by only a few kelvin.

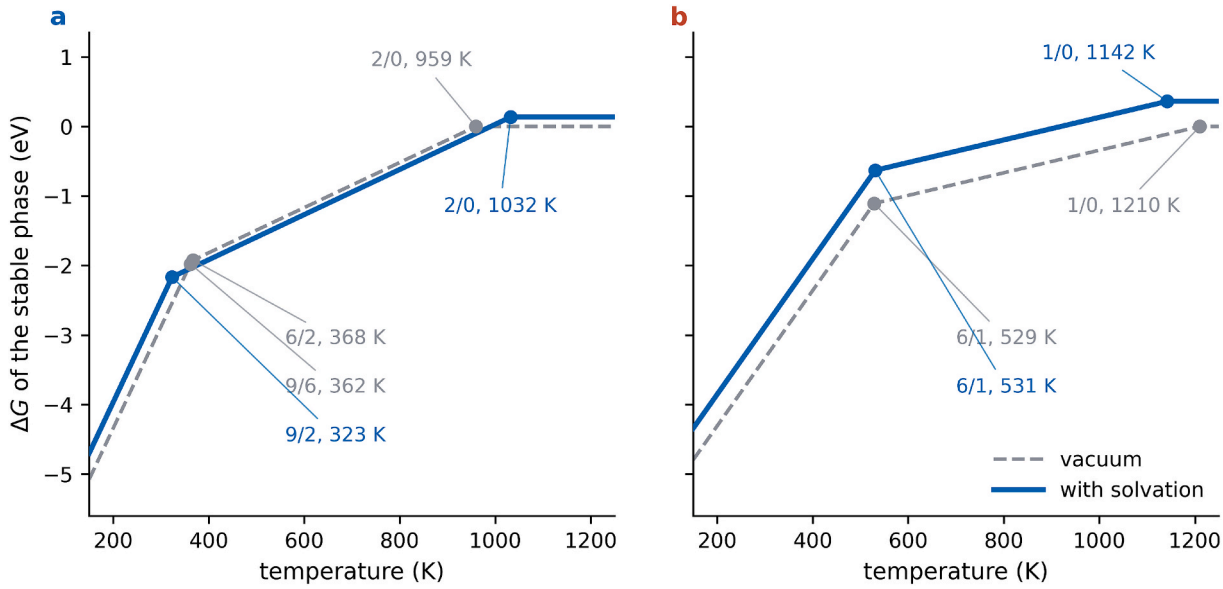


Fig. 8. Free energy of the most stable phase as a function of temperature for the pristine (a) and vacancy (b) supports, both in vacuum (dashed line) and when the interface is solvated by liquid water (solid line). X/Y marks the positions of the phase transitions between a structure with X water molecules adsorbed on the cluster and a structure with Y water molecules (not counting any additional MM water molecules from the solvation calculation). Pressure is the vapor saturation pressure of water throughout.

4. Conclusions

Treating the $Ru_{10}/TiO_2(1\,0\,1)$ interface as a fluxional sub-nanometer cluster equilibrated with liquid water, rather than considering just a flat metallic Ru surface, qualitatively changes the predicted active state of the catalyst. Grand-canonical sampling identifies four features of that state that together close the gap between flat-Ru thermodynamics and the OH-mediated chemistry observed on working Ru/TiO_2 catalysts.

(i) Hydroxyl is the dominant oxygenated adsorbate across all coverages on both pristine and oxygen-vacancy supports (44.9% and 50.2% of all oxygenated adsorbates), and it sits at single-Ru top atoms at the cluster periphery (89% pristine, 82% vacancy). In direct contrast to flat Ru surfaces, this predicts that OH is a stable surface species at catalytic temperatures in aqueous conditions.
(ii) The thermodynamic phase diagram on the vacancy support carries an intermediate-coverage 6-water plateau (enriched in OH and absent from the pristine phase diagram) across the 400–530 K window relevant to operando hydrodeoxygenation.
(iii) Bader analysis of the eight phase-diagram global minima shows per-Ru charge rising by ∼+0.18 e per Ru–O contact. The bare-cluster total charge on the vacancy ∼ 0.9 e lower than on pristine, consistent with partial transfer of vacancy-liberated polaronic electrons into the Ru cluster. The OH species carry a net charge of ∼ − 0.5 e per unit: typically they are a partially anionic σ-bonded hydroxyl on a Lewis-acidic Ru top atom.

Together these findings reconcile the apparent contradiction between flat-Ru aqueous thermodynamics, which predicts no stable OH at catalytic temperatures [3], and the experimental persistence of OH-mediated chemistry on working Ru/TiO_2 catalysts, by identifying the 6-water OH-rich phase as the relevant active state for aqueous-phase upgrading. More generally, they argue that mechanistic studies of supported sub-nanometer cluster catalysts cannot be reduced to single-crystal surface analogies: the active site is the cluster periphery, OH is recovered as a major adsorbate species (in contrast to the fully dissociated O + H of the flat-Ru reference), and the support defect is an electronic, not merely geometric, participant in the chemistry (by injecting negative charge into the cluster and enabling the formation of hydridic H atoms at the triple phase boundary).

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CRediT authorship contribution statement

Joshua M. Sims: Writing – review & editing, Writing – original draft, Visualization, Formal analysis, Data curation, Conceptualization. **Carine Michel:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article (detailed sampling workflow, discussion of sampling sufficiency, supplementary figures, description of Solvhybrid and structures of various global minima) can be found online at https://doi.org/10.1016/j.apsusc.2026.168653.

Data availability

All data has been made available in the submission

The full DFT sampling dataset (1774 relaxed structures), together with the relabeling pipeline, is available on Zenodo at https://doi.org/10.5281/zenodo.20179039. Further data supporting this article, including full computational details, sampling-sufficiency diagnostics, Bader-charge tables, and atomic coordinates (.xyz) of the eight phase-diagram global minima, have been included as part of the Supplementary Information (SI).

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